Agile Software Development and DevOps Lab

Experiment 6

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Batch	: T22	**Subject**	: ASD&D

Aim: To study and implement the Agile software project lifecycle using Jira with DevOps pipeline integration.

Theory

What is Jira? Jira is an Agile project management and issue-tracking tool from Atlassian. Teams use it to plan sprints, track tasks and bugs, and follow releases through customizable workflows.

Core concepts

- **Project / space:** a shared workspace for a product or team.
- **Issue hierarchy:** an Epic is a large goal, a Story is a feature written from the user's point of view, a Task or Bug is a technical item or defect, and a Sub-task is a smaller step inside one of these.
- **Backlog:** the prioritized list of work that has not yet been scheduled.
- **Sprint and Scrum board:** a time-boxed cycle (usually 1–2 weeks) in which issues move across columns such as To Do, In Progress, In Review and Done.
- **Story points:** relative estimates of the effort, complexity and risk of an issue.

Deployment models. Jira Cloud is hosted by Atlassian and reached through a browser at a site such as *name*.atlassian.net, with no server to maintain. Jira Data Center is self-managed on private or cloud infrastructure for organizations that need local data control and high availability.

DevOps integration.

- **Version control link:** apps such as GitHub for Jira connect a repository to a Jira site.
- **Issue keys:** mentioning a key (for example SCRUM-4) in a branch name, commit message or pull request title ties that code activity to the Jira issue.
- **Automation rules:** event-driven rules can move an issue to In Review when a pull request is opened and to Done when it is merged.

Procedure

- The project used for this experiment is **IdeaForge**, a web application for collaborative brainstorming in shared rooms (see Figs. 7 and 8).
- Created a Jira Cloud account and chose **Software development** as the type of work, then selected how the team plans to use Jira.
- Created a Scrum space named **IdeaForge** with the workflow statuses To Do, In Progress, In Review and Done, and added issues for the IdeaForge web app (for example "add AI idea generation", "filter ideas", "edit room names" and "whiteboard creation").
- Used the sprint board to move issues through the workflow columns.
- Connected the GitHub account **Aaditya179** to Jira using GitHub for Jira, limited to selected repositories.

- In the IdeaForge repository, created the branch **IDEAF-1-search**, committed a README change with the message "IDEAF-1: update README", and pushed it to GitHub.

Output

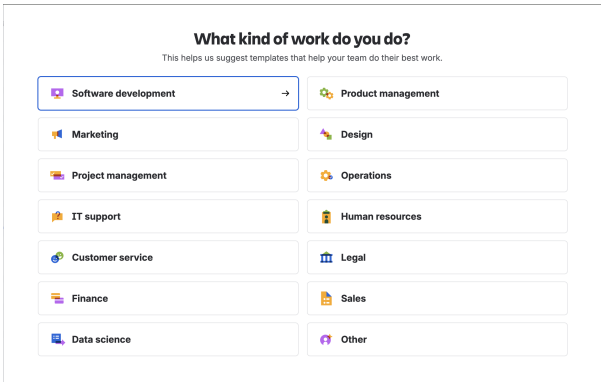


Fig. 1 — Jira onboarding — selecting "Software development" as the type of work

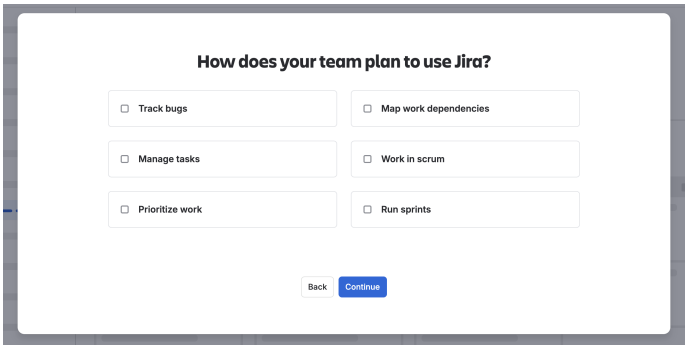


Fig. 2 — Jira onboarding — "How does your team plan to use Jira?"

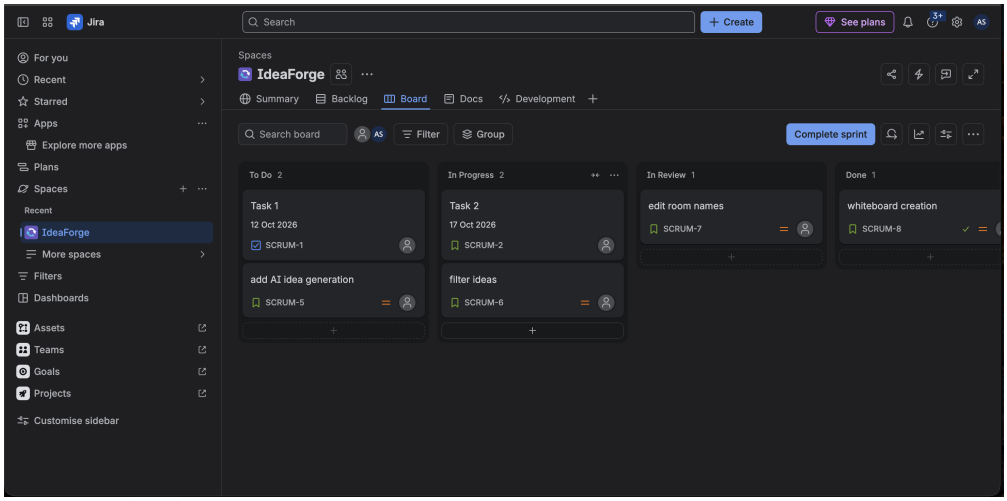


Fig. 3 — IdeaForge Scrum board with To Do, In Progress, In Review and Done columns

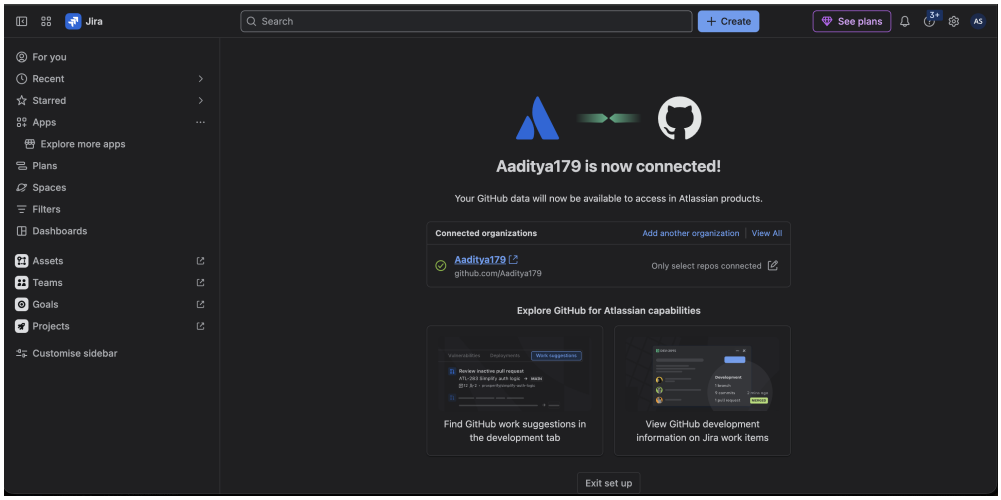


Fig. 4 — GitHub for Jira — the account Aaditya179 is connected to Jira

```
(base) adityasurve@Adityas-MacBook-Air IdeaForge-main %    git checkout -b IDEAF-1-search
   git add README.md
   git commit -m "IDEAF-1: update README"
   git push -u origin IDEAF-1-search
Switched to a new branch 'IDEAF-1-search'
[IDEAF-1-search (root-commit) 0d2e2a1] IDEAF-1: update README
 Committer: Aaditya Surve <adityasurve@Adityas-MacBook-Air.local>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly:

    git config --global user.name "Your Name"
    git config --global user.email you@example.com

After doing this, you may fix the identity used for this commit with:

    git commit --amend --reset-author

 1 file changed, 71 insertions(+)
 create mode 100644 README.md
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 1.58 KiB | 1.58 MiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
remote:
remote: Create a pull request for 'IDEAF-1-search' on GitHub by visiting:
remote:      https://github.com/Aaditya179/IdeaForge/pull/new/IDEAF-1-search
remote:
To https://github.com/Aaditya179/IdeaForge
 * [new branch]      IDEAF-1-search -> IDEAF-1-search
branch 'IDEAF-1-search' set up to track 'origin/IDEAF-1-search'.
(base) adityasurve@Adityas-MacBook-Air IdeaForge-main %
```

Fig. 5 — Terminal — creating branch IDEAF-1-search, committing and pushing to GitHub

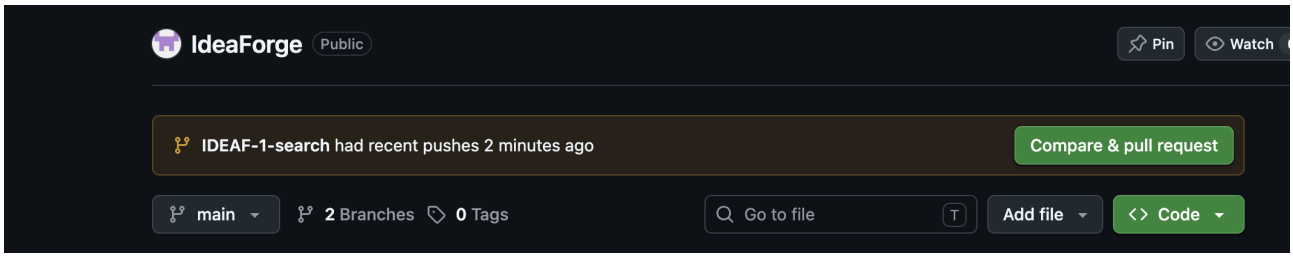


Fig. 6 — GitHub — IdeaForge repository showing the pushed branch with the Compare & pull request option

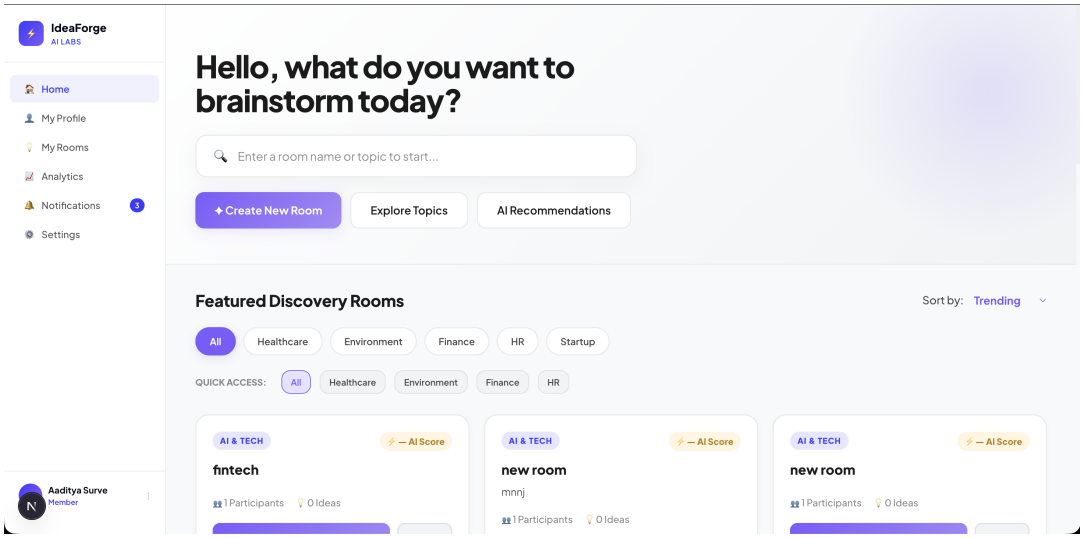


Fig. 7 — IdeaForge web application — home page with discovery rooms

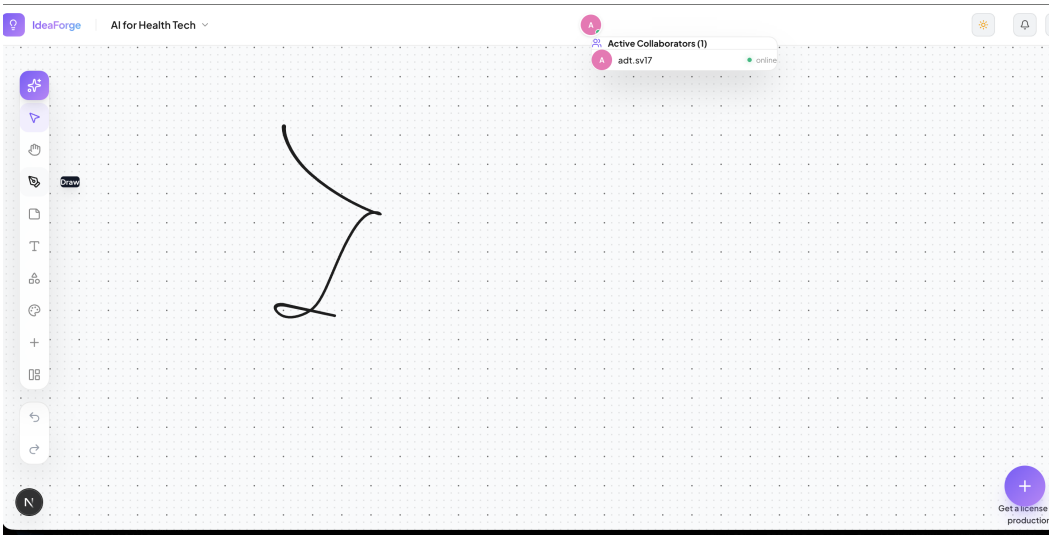


Fig. 8 — IdeaForge brainstorming room (AI for Health Tech) with the drawing canvas and active collaborators

CONCLUSION

In this experiment, we set up Jira Cloud, created a Scrum space for the IdeaForge project with a board of To Do, In Progress, In Review and Done columns, and tracked work items on it. We connected a GitHub account to Jira using GitHub for Jira and created a branch and commit whose names carry an issue key, which is the basis for linking code activity to Jira issues in an Agile and DevOps workflow.